

Polymers

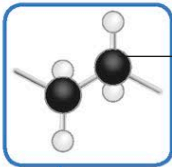
Some molecules join together in a chain to form long polymers (meaning “many parts”). The smaller molecules that make up the polymer are called monomers. There are many important polymers in living things. Cellulose, which makes up wood, is the most abundant natural polymer on Earth. The DNA in our bodies, and starch in foods such as pasta, rice, and potatoes are also polymers. Polymers can be man-made, too. Synthetic polymers include a vast array of different plastics.

Plastic polymers and recycling

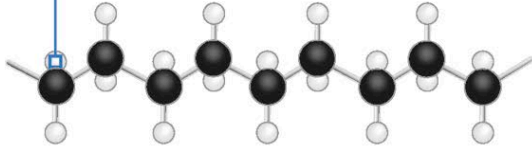
The first man-made polymers were attempts to reproduce the natural polymers silk, cellulose, and latex (see pp.58–59). Today, plastics play a massive role in the way we live, but they also pose a serious risk to the environment. In 1988, an identification code was developed to make plastic recycling easier. The code’s symbols let the recyclers know what plastic an object is made of, which matters when it is time to process and recycle it.

What makes a polymer

A polymer is like a long string of beads, with each bead, or monomer, in the string made up of exactly the same combination of atoms. Shorter ones, with just two monomers, are called dimers, while those with three are known as trimers.

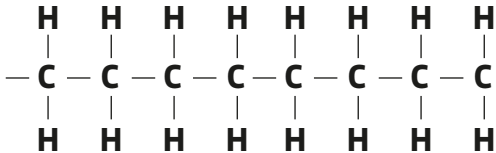


The monomer ethene is made up of two carbon atoms and four hydrogen atoms.



Polyethylene polymer

A string of ethene monomers is known as polyethylene (or polyethene/polythene). There are several thousand ethene monomers in a polyethylene polymer.



Type of plastic	Symbol	Properties	Use
Polyethylene terephthalate	 PET or PETE	Clear, lightweight but strong and heat-resistant. Good barrier to gas, moisture, alcohol, and solvents.	• Water bottles • Food jars • Ovenproof film
High-density polyethylene	 HDPE	Tough; can be stretched without breaking, and easy to process. Resistant to moisture and solvents.	• Milk containers • Trash cans with wheels • Juice bottles
Polyvinyl chloride	 PVC	Strong; resistant to chemicals and oil. Rigid PVC is used in construction; flexible PVC in inflatables.	• Pipes • Toys and inflatables • Flooring
Low-density polyethylene	 LDPE	Flexible and tough, can withstand high temperatures. Good resistance to chemicals. Easy to process.	• Plastic bags • Snap-on lids • Six-pack rings
Polypropylene	 PP	Tough, flexible, and long lasting. High melting point. Resistant to fats and solvents.	• Hinges on flip-top lids • Plastic medicine bottles • Concrete additives
Polystyrene	 PS	Can be solid or foamed. Good for insulation and easy to shape, but slow to biodegrade.	• Disposable foam cups • Plastic cutlery • Packaging
Miscellaneous	 Miscellaneous	Other plastics such as acrylic, nylon, polylactic acid, and plastic multi-layer combinations.	• Baby bottles • Safety glasses • “Ink” in 3-D printers